SUNG-JOO LEE

+1 332 258 6675
sungjoolee2@gmail.com

EDUCATION

Columbia University, New York, USA — September 2022 - Present
PhD Candidate in Ecology, Evolution, and Environmental Biology
Dissertation (tentative): Effect of resource landscape on deer-mediated pathogen transmission in human-dominated landscapes at multiple hierarchical levels
Advisor: Prof. Maria Diuk-Wasser

Korea University, Seoul, South Korea — March 2015 - June 2021
MSc & BSc in Environmental Science & Ecological Engineering
Major in Environmental Planning and Landscape Architecture
Master's thesis: Path-Finding Algorithm as Dispersal Assessment Method for Invasive Species with Human-vectored Long-distance Dispersal Event - A case study on pinewood nematodes
MSc Advisor: Seongwoo Jeon
Overall GPA: MSc 4.31/4.5 (4.0/4.0)
BSc 4.04/4.5 (3.75/4.0)

RESEARCH INTERESTS

Disease ecology, Zoonotic diseases, Movement ecology, Spatial modeling, Landscape ecology, and Structured decision-making

RESEARCH EXPERIENCE

Columbia University — US
Doctoral Researcher — *October 2022 – Present*

- Evaluate the effect of feeding sites on white-tailed deer behavior and contact rates using a fine-scale GPS trajectory dataset
- Estimate host species/community space use patterns via camera trapping and simulate the cost-effective management scenarios for tick-borne diseases
- Organize and conduct fieldwork, including tick dragging, mouse trapping, camera trapping, deer sampling, and deer tracking

Korea Environment Institute — Sejong, South Korea
Researcher — *July 2021 – June 2022*

- Conducted a review on home range estimation methods of wild animals
- Managed a large tracking dataset (∼200 individuals of Black-tailed gulls (*Larus crassirostris*)) and used them to investigate the ecological impact of offshore wind farms on seabirds

Korea University — Seoul, South Korea
Undergraduate and Master's Researcher — *September 2019 – June 2021*

- Investigated the optimal management strategy for pinewood nematodes using multiple spatially explicit removal strategies under climate change scenarios
- Developed a framework of using a path-finding algorithm to evaluate human-mediated dispersal of pine wood nematodes (*Bursaphelenchus xylophilus*) vectored by pine beetles

- Collected *Galloisiana odaesanensis* and *Galloisiana odaesanensis* from its natural habitat and conducted a literature review to support its registration on the IUCN Red List

LEADERSHIP EXPERIENCE

Columbia University
Teaching Assistant *January 2023 – May 2024*

- Assisted in teaching three undergraduate/graduate courses ranging in size from 5 to 200 students. Topics included: intermediate statistics for ecology and evolutionary biology, behavioral biology of primates, and science for sustainable development
- Led weekly TA sessions where I lectured on R coding and/or led discussions on lecture materials
- Prepared course material including lectures, exams, and homework
- Supervised students in final projects, graded exams, and weekly homework

Korea University September 2019 - June 2021
Teaching Assistant & Program Organizer

- Tutored 12 sophomores in GIS and R skills required for undergraduate team projects for a semester
- Facilitated a remote sensing workshop for the International Society of Photogrammetry and Remote Sensing (ISPRS), organizing 10 lecture sessions, hands-on sessions, and a field trip

PUBLICATIONS

1. Lilly, M.V., Davis, M., Kross, S.M., Konowal, C.R., Gullery, R., **Lee, S.-J.**, Poulos, K.I., Gregory, N., Nagy, C., Cozens, D.W., Brackney, D.E., Fernandez, M.P., Diuk-Wasser, M., 2025. Functional connectivity for white-tailed deer drives the distribution of tick-borne pathogens in a highly urbanized setting. *Landscape Ecology* 40(5): 87.

2. **Lee, S.-J.**, Lee, W.-S., 2022. Animal Home Range Estimators: A Review. *Korean Journal of Environment and Ecology* 36(2): 202-216. [In Korean with English Abstract]

3. **Lee, S.-J.**, Cho, H., Choi, Y., Choi, W.I., Chung, H., Lim, N.O., Nam, Y., and Jeon, S.W., 2022. Path-finding Algorithm as a Dispersal Assessment Method for Invasive Species with Human-vectored Long-distance Dispersal Event. *Diversity and Distributions* 28(6): 1214-1226.

4. Lim, N.O., Hwang, J., **Lee, S.-J.**, Yoo, Y., and Jeon, S.W.,2022. Spatialization and Prediction of Seasonal NO_2 Pollution Due to Climate Change in the Korean Capital Area through Land Use Regression Modeling. *International journal of environmental research and public health* 19(9): 5111.

5. Lee, K.I., **Lee, S.-J.**, Tuvshinjargal, N., Lee, E.S., Lee, G.G., and Jeon, S.W., 2020. Urban Heat Vulnerability assessment – Case Study on Suwon, Korea. *Journal of Climate Change Research* 11(6-1): 629-641. [In Korean with English Abstract]

6. **Lee, S.-J.**, Ryu, J.E., and Jeon, S.W., 2020. An Analysis of Environmental Policy Effect on Green Space Change using Logistic Regression Model: The Case of Ulsan Metropolitan City. *Journal of the Korean Society of Environmental Restoration Technology* 23(4): 13-30. [In Korean with English Abstract]

7. Chung, H.-Y., Lee, Y-W, Park, S.-H., Lim, C., Park, S.-H., Hong, M.-H., Lee, Y.-S., Lee, D., Shin, H., **Lee, S.-J.**, Oh, H.-K. & Gwon, S. 2018. Galloisiana kosuensis. The IUCN Red List of Threatened Species 2018: e.T113551876A113555352.

8. Chung, H.-Y., Lee, Y-W, Park, S.-H., Lim, C., Park, S.-H., Hong, M.-H., Lee, Y.-S., Lee, D., Shin, H., **Lee, S.-J.**, Oh, H.-K. & Gwon, S. 2018. Galloisiana odaesanensis. The IUCN Red List of Threatened Species 2018: e.T113552199A113555367.

(Ongoing)

Wu, L., **Lee, S.-J.**, Kopsco, H.L., Lee, A., Huang, Y., and Diuk-Wasser, M., *in revision.* Integrating environmental and economic decision-making frameworks for household-level tick-borne disease management. *Emerging Infectious Diseases.*

Fu, W., Lilly, M.V., **Lee, S.-J.**, Kopsco, H., Surasinghe, T., Fernandez, M.D.P., Popescu, V., Stark, J., Edwards, J., Gould, H., Kelly, P.H., and Diuk-Wasser, M., *in review.* Mapping tick-borne hazards across gradients of urban intensity in metropolitan regions. *Parasites & vectors.*

SKILLS

Skills	Geospatial analysis (R, QGIS, ArcGIS) Biostatistics, Data wrangling (Python, R) Remote sensing (Google Earth Engine, ERDAS imagine) LaTeX
Languages	Korean (Native) English (Fluent)

SCHOLARSHIPS

Korea University	Veritas Program Scholarship (2018) – \$1,000 Special Scholarship (2018) – \$1,800 (Half tuition) : Equivalent to Academic Excellence Scholarship Academic Excellence Scholarship (2015) – \$1,800 (Half tuition)
Columbia University	PhD Full-Package Fellowship (2022–2027)